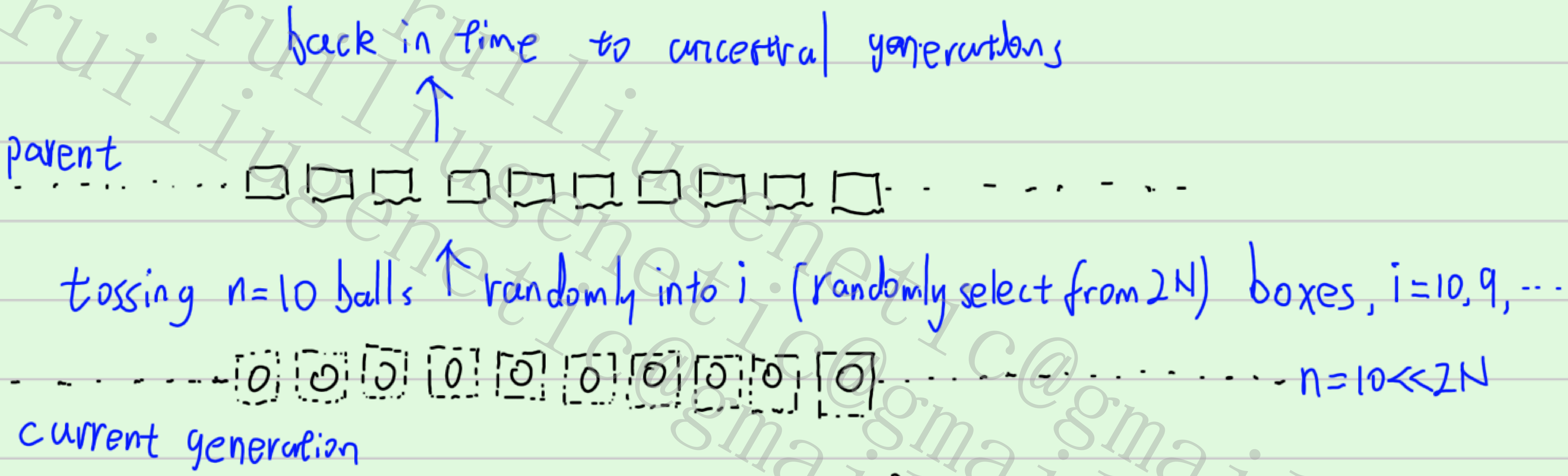
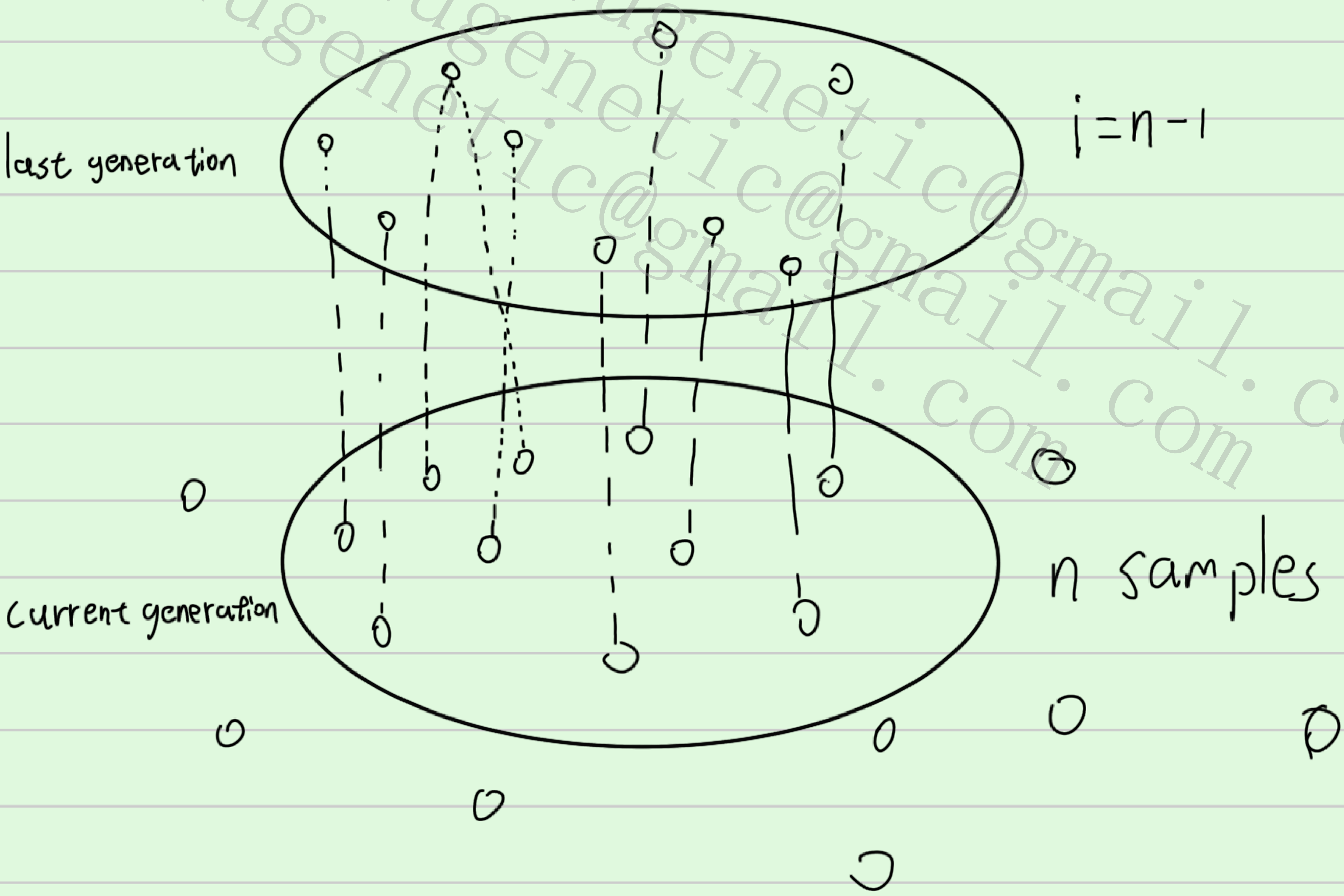


$$G_{n,i} = \frac{S_n^{(i)} N_{[i]}}{N^n} \begin{cases} G_{n,n} = 1 - \frac{C_n^2}{2N} + o\left(\frac{1}{N}\right), & i=n \\ G_{n,n-1} = \frac{C_n^2}{2N} + o\left(\frac{1}{N}\right), & i=n-1 \end{cases}$$

$$\xRightarrow{N\to\infty} G_{n,n} + G_{n,n-1} = 1$$

N generation as unit : $G_{n,n-1} \rightarrow$



1. 溯祖事件

当$N\to\infty$，每代溯祖($G_{n,n-1}$)或不溯祖($G_{n,n}$)视作Bernoulli trials

X	coalescent	not
p	$1-C_n^2/2N$	$C_n^2/2N$

(单倍体为N)

t代发生k次的概率

$$P_t\{K=k\} = C_t^k (G_{n,n-1})^k (G_{n,n})^{t-k} \xrightarrow{\lambda = tG_{n,n-1}} \frac{\lambda^k}{k!}e^{-\lambda}$$

coalecent事件的等待时间

$$P\{T=t\} = G_{n,n-1}(1-G_{n,n-1})^t \xrightarrow[\text{概率为}\lambda\Delta t]{\Delta t\text{内发生}} \lambda\Delta t(1-\lambda\Delta t)^{t/\Delta t}$$

根据$x=0$处 e^{-x}的Taylor展开，当x很小时 $e^{-x}\approx 1-x$

得到 n个样本溯祖一次的等待时间 T_n的密度函数为

$$\xrightarrow{\Delta t\lambda e^{-\lambda\Delta t\cdot t/\Delta t}} f_{T_n}(t)\,dt = \lambda e^{-\lambda t}\,dt \qquad \left(\lambda = \frac{C_n^2}{2N}\right)$$

以2N代为单位时间，$\lambda = C_n^2$

$$E[T_n] = \frac{1}{\lambda} = \frac{2}{n(n-1)}, \quad Var[T_n] = \frac{1}{\lambda^2}$$

n个样本的总溯祖时间(i.e. 树高) $T_{MRCA} = T_n + T_{n-1} + \cdots T_2$. 且各次溯祖事件相互独立

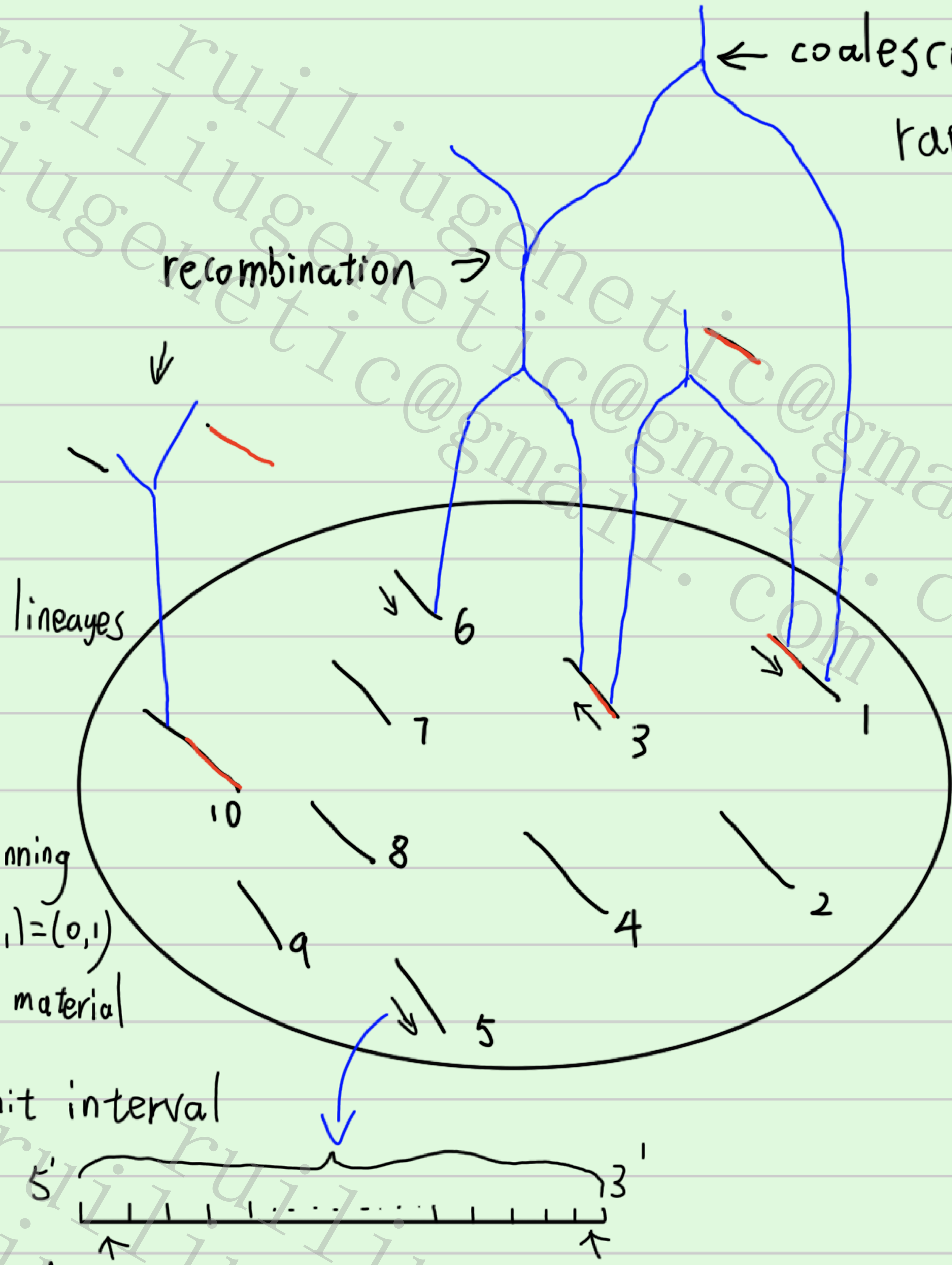
$$E[T_{MRCA}] = \sum_{i=2}^{n} E[T_i] = \frac{2}{n(n-1)} + \frac{2}{(n-1)(n-2)} + \cdots = 2\left(1-\frac{1}{n}\right)$$ for both W.F & Moran model

$$E[T_{total}] = 2\sum_{i=2}^{n} i\,\frac{2}{i(i-1)}$$

rate $I_{ij}=1 \xrightarrow[\text{标准}]{\text{对应}} C_k^2$, $k=2$

瞬时溯祖率 $\lambda_c = \sum_{i\neq j} I_{ij}$

k ancestral lineages



k=10 at beginning

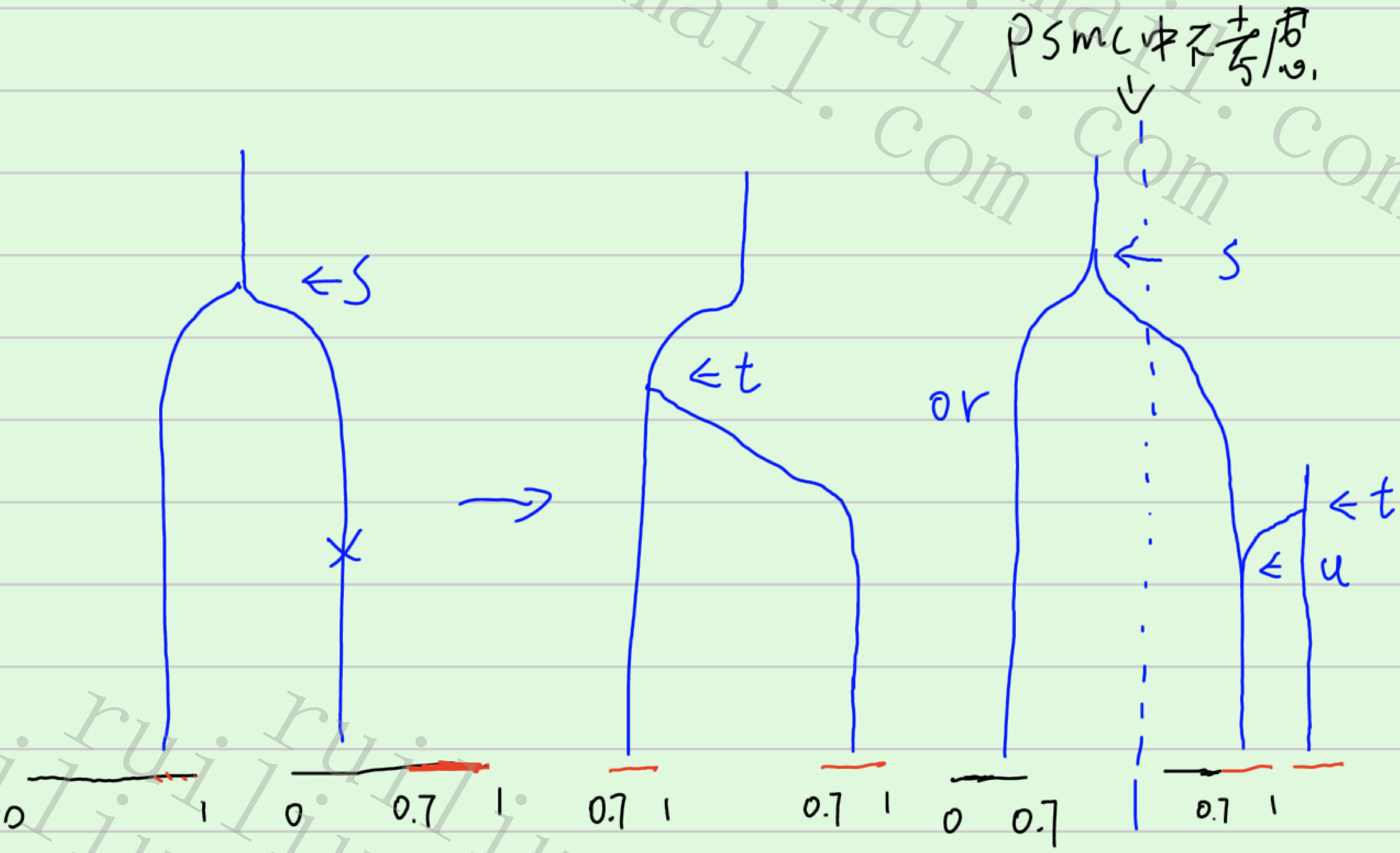
$m_i=1$, $(x_{i1}, y_{i1})=(0,1)$ for all k material

$X_i = \{(x_{i1},y_{i1}),(x_{i2},y_{i2}),\dots,(x_{im},y_{im})\}$

after coalecence, if any interval remain just one ancestral lineage i.e. only the MRCA of that interval, k=1 for the interval, remove the interval.

(若某区间在溯祖之后，所有样本合为一个MRCA，则移除该区间)

禁止无共祖片段之间溯祖，即 $I_{ij}=0$，如果两片段间无重叠ancestral片段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